Mini Project Report on

IOT BASED RAIN PROTECTION SYSTEM

Submitted by

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In partial fulfillment for the award of degree of

BACHELOR OF TECHNOLOGY

In

ELECTRICAL AND ELECTRONICS ENGINEERING

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

INSTITUTE OF ENGINEERING AND TECHNOLOGY

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CERTIFICATE

This is to certify that the mini project report entitled **"IOT BASED RAIN PROTECTION SYSTEM"** *is a bona fide record of the work done by* **Ms. ISHITA V.K [IEASEEE020], Mr. K.ANSHUL [IEASEEE021], Mr. KIRAN JOY.T [IEASEEE022], Mr. MOHAMED RIZWAN [IEASEEE023]** *under our supervision in fulfillment of the requirement the award of degree of bachelor of technology in electrical and electronics engineering from Calicut University for the year 2020-2021*

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We express our gratitude to **Dr. RANJITH C**, Principal, Institute of Engineering and Technology, University of Calicut and **Mr. SREEJITH C.S** , Head of the Department, Dept. of Electrical and Electronics Engineering, for their valuable suggestions and constantly inspiring during the course of this work.

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Finally, we extend our sincere thanks to our dear parents and friends who helped us to build up confidence.

ABSTRACT

In this paper we propose a model which can be highly beneficial in household and agricultural activities especially for farmers and industries. This project is very relevant in southern part of India where sudden rains can be expected during the months of April to May.

Majority of spices and virgin oils in India are produced in the southern states like Kerala, TamilNadu and Andrapradesh. During the process of manufacturing, the raw materials like Chilli, Pepper, Turmeric, Coriander, Coconuts needs to be sun dried to improve its quality and shelf life. Most of these materials are sun dried outdoors and sudden rains can completely ruin these materials. Therefore human supervision is necessary to ensure that these materials are taken indoors at the right time so as to protect them from rain. Since these materials are always dried in a bulk amounts a single human error can result to a huge margin of loss.

Here is where our project comes into the scene. The main theme of this project which is presented in the paper is Automation. We intend to help farming households and big industries to protect their raw material from sudden rain and eliminate regular human supervision by automizing the project.

The project involves a rain sensor which alerts the user when it rains. The user can protect the material using a rain shed which can be controlled using a mobile app-”Rain guard” at the comfort of their homes. A notification is immediately sent to the mobile phone of the user when slight amount of rain is sensed. The user just has to slide a slider provided in the mobile app to turn on the protection. This app is a life saver as it also lets the user to monitor the real time temperature, humidity and light levels which to some extent helps in predicting sudden rains. The rain levels can also be monitored in this app which the user to know the intensity of rain. This project is ideal to protect the materials from low to medium amounts of rain. If the rain is very heavy, then it is recommended to get back the material indoor.

When the rain stops, the user can turn off the protection and again resume the process of drying. The equipment used in the project is cost effective and highly accurate. Hence is can be affordable for all class of people. The components and sensors used in the project will be discussed in detail as you go through this paper. Marketing this project can definitely improve the quality of the materials to some extent by reducing some possible human errors. The applications of this project can be broadened to various other sectors which requires their materials to be protected from rain. In a world where technology rapidly grows day by day, we hope that this project can simplify activities in lives of many people and can be adapted to various activities as per the requirement.

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CHAPTER 1

INTRODUCTION

As human beings we cannot control the natural phenomenon such as rain, humidity, high temperature, etc. Some of the measures are taken against this environmental hazard but they are performed manually. Here comes the need of automation. Automation greatly decreases the need for human sensory and mental requirements as well. An automation system consisting of a connection between hardware and software has freed the individuals from their day to day chores. In this paper we try to establish new intelligent system which helps to protect the user daily home application and other useful material against environmental impact like rain.

This paper is based on the project which is an embedded system consisting of a weather station monitoring and rain protection using NodeMcu ESP8266, DHT11 humidity and temperature sensor, LDR sensor, SG90 micro servo motor and FC37 rain sensing system to protect materials from rain. NodeMcu is the processing and controlling unit of this system which receives and processes the data from the sensors. The WIFI is an interface between NodeMCU and user's mobile and is responsible for communication between them. The mobile phone App can be used as a controller within the Wi-Fi frequency range (2400-2484MHz). If the WIFI network is available the user can switch on/off the system and to receive the alert messages. The whole sensors will be placed at open place. They will continuously send the real time light levels, temperature, humidity values to NodeMCU. If the rain falls, then the NodeMCU will send an alert message to the mobile station and roof will get open when a slider dragged and also we can close the roof using the mobile app. Hence there exists an electro-mechanical system which continuously monitors the rain and closes the roof by a slider in the mobile app to protect things from wet.

CHAPTER 2

OBJECTIVE

The main objective of this project is protect objects (clothes or any other kind of raw materials like coconut, chilli etc.) from sudden and unexpected rains. We aim-

- **To detect small to high levels of rains using rain sensor.**

Detection of rain is very simple. The output of the rain sensor is communicated to NodeMcu ESP 8266 via programmed instructions using Arduino IDE.

- **To protect material using a rain shed.**

The shed works with help of an micro servo motor. Once a small to medium amount of rain is sensed in rain sensor, the user slides the micro servo motor to 180degrees ,thereby opening the shed and protecting our material. When the rain sensor is dry, the user can slide back the shed to the original position.

- **To alert the user about rain using a mobile app.**

The mobile app notifies the user about the rain and asks the user to open the shed with help of the app. The user can simply use the slider option in the app to open or close the shed. The mobile app also updates the user about the real time temperature, humidity and the light levels

CHAPTER 3

LITERATURE SURVEY

3.1 An Internet based wireless home automation system for multifunctional devices

The aim of home automation is to control home devices from a central control point. In this paper, we present the design and implementation of a low cost but yet flexible secure internet based automation system. The communication between the devices is wireless. The protocol between the units in the design is enhanced to be suitable for most of the appliances. The system is designed to be low rest and flexible with the increasing variety of devices to be controlled.

3.2 Design of a tropical rain- Disaster alarm system: A new approach based on wireless sensor networks and acoustic rain rate measurements

This paper discusses the design of a broadband wireless network infrastructure which itself is a rain measurement platform for applications such as disaster alarm and sudden.Hazard decision management systems. A sensor Test bed is setup which consists of a hybrid broadband wireless network in conjunction with real-time acoustic rain rate point sensors and complementary rain gauges. The test bed simulates the commercial deployment of a line-of sight wireless backbone (implemented via a 26 GHz line of sight link) and broadband wireless access network at 5 GHz and 24 GHz. Combined wireless signal fade, acoustic power and tipping bucket rain rate measurements over a several month span indicate the feasibility of using rain-induced attenuation and fade durations to trigger imminent-hazard alerts.

3.3 Rainfall monitoring using acoustic sensors

This study is about the design, development, and field testing of acoustic sensors for rain measurements. An Android-based acoustic sensor is designed and tested. The sensor can upload data files to a web server, and can trigger an SMS alarm when rainfall data exceeds safety thresholds. An Arduino-based acoustic sensor is also designed and tested, which is an alternative, low-cost acoustic sensor for rainfall. Rainfall data from the sensors are graphed, analyzed, and compared vis-à-vis to data from tipping bucket rain gauges. We report, for the first time, the performance of an acoustic rain sensor and a tipping bucket in the same device integrated with an Android phone. This new configuration shows the difference between a real-time acoustic rain sensor and a tipping bucket, which is an accumulation sensor. The analysis features and performance of the acoustic sensors directs to the development of low-cost devices for gathering rain data, which can supplement standard rain measurement devices.

CHAPTER 4

BLOCK DIAGRAM DESCRIPTION

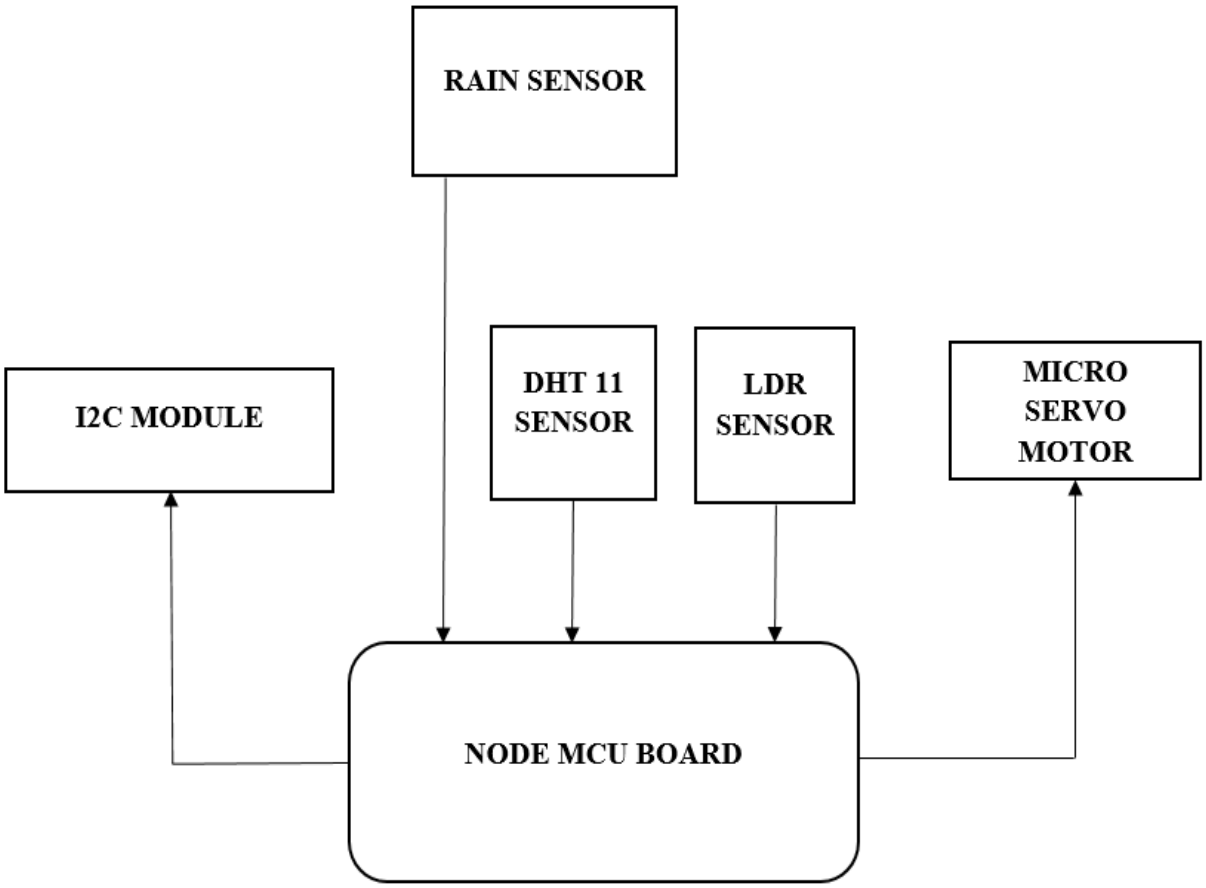


Figure 1. Block Diagram of IOTRPS

The block diagram of the IOT based Rain protection system is shown in the Fig. . This block diagram depicts the sensors and other components used in this project. The core of this project is NodeMCU ESP8266 Wi-Fi development board which acts as the micro controller. All other sensors and components are interfaced to this micro controller.

Three main sensors used in this project are the Rain sensor, the DHT11 sensor and the LDR sensor. The Rain sensor senses the presence of rain, the DHT11 sensor senses the real time temperature and humidity and the LDR sensor senses the light levels near the premises of the project. All three sensors sends back the output to the micro controller board. Two other components used in this project is the L2C Display module and the micro servo motor.

The micro servo motor is the most important component of the project. This component acts as the Rain shed. Whenever rain is sensed in the rain sensor the servo motor rotates the rain shed towards the object thereby protecting it from rain. The L2C module displays the real time temperature, humidity, light levels and the rain levels.

CHAPTER 5

CIRCUIT DIAGRAM EXPLANATION

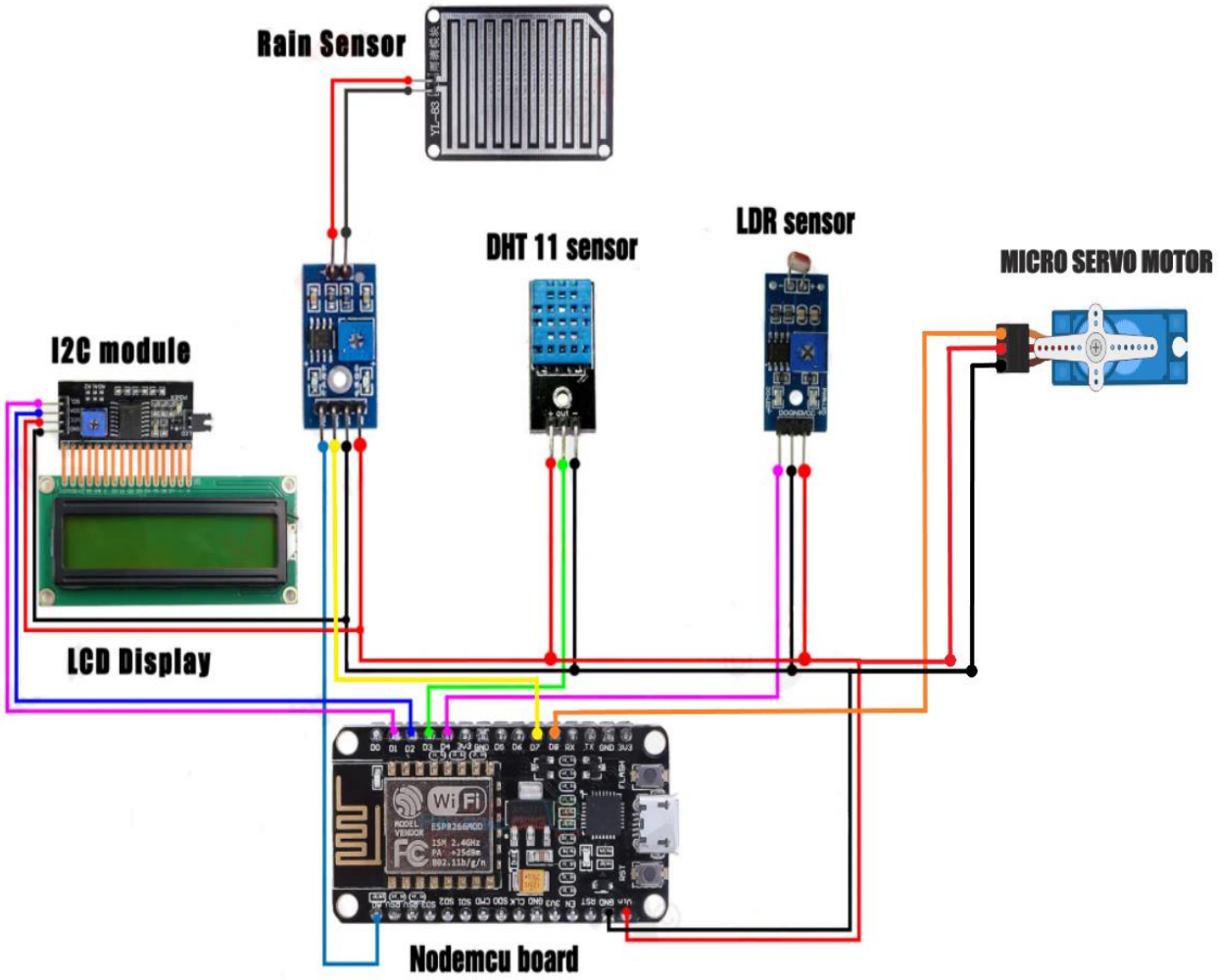


Figure 2. Circuit Diagram of IOTRPS

The circuit diagram of the IOT based Rain Protection system is shown in the Fig. . It shows the microcontroller NodeMcu ESP8266 Wi-Fi dev. Board, I2C display module, Micro servomotor SG90, FC37 Rain Sensor, LDR Sensor, DHT11 Sensor.

The Voltage and Ground pins of each of the components are commonly connected to the VCC and GND pins of the NodeMcu respectively via breadboard. The SCL and SDA pins of the I2C pins are connected to the D1 and D2 pins of the NodeMcu. These pins are specifically reserved for the purpose of display pins. The PWM pin of the servomotor is connected to the D8 terminal of NodeMcu. The digital pins of the DHT11 sensor and the LDR sensor and connected to the D3 and D4 pins of the NodeMcu respectively. The Analog and digital pins of the rain sensor is connected to the A0 and D7 terminals of the micro controller board.

CHAPTER 6
INTERFACING OF SENSORS AND COMPONENTS WITH NODEMCU ESP8266

INTERFACING RAIN SENSOR WITH NODEMCU

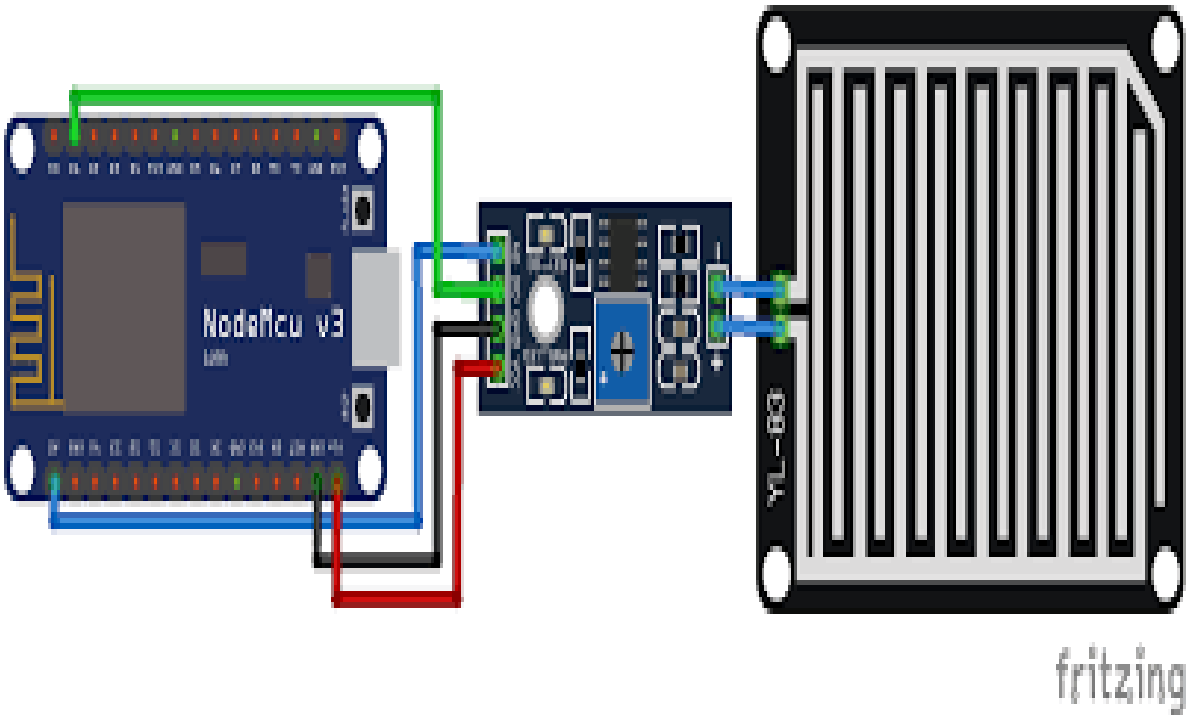


Figure 3. Interfacing FC37 with NodeMcu

The FC37 rain sensor sends both digital and analog values as output to the NodeMcu ESP8266. FC37 has 4 pins namely, VCC,GND,D0,A0. The VCC and GND are connected to the Vin and GND of the NodeMcu. The digital pin D0 of FC37 is connected to D7 of NodeMCU and the analog pin A0 of FC37 is connected to A0 of the NodeMCU. The intensity of the rain is communicated through the analog pin whereas the present state of the sensor(dry or wet) is communicated through the digital pin.

INTERFACING DHT11 SENSOR WITH NODEMCU

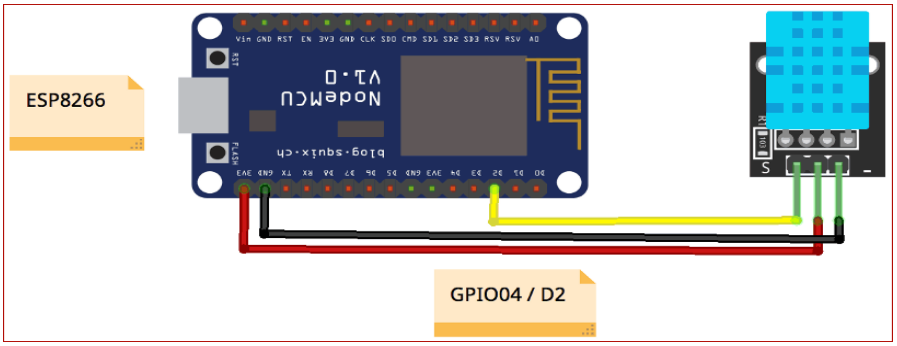


Figure 4. Interfacing DHT11 with NodeMcu

The DHT11 sensor provides the real time temperature and humidity near the project. This is more of an additional feature and does not have any direct role in protecting the project.The DHT11 sensor has '+','-',OUT pins. The '+' and '-' pins are connected to the Vin and GND pins of the NodeMCU and the OUT pin of DHT11 sensor is D3.

INTERFACING LDR SENSOR WITH NODEMCU

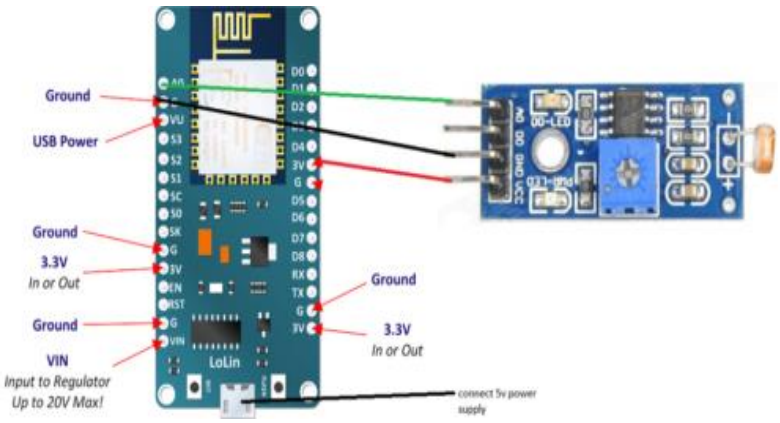


Figure 5 Interfacing LDR with NodeMcu

The LDR sensor senses the real time light levels near the project and reports back the output as high or low. The LDR has three terminals, D0, VCC and GND. The VCC and GND are connected to the Vin and GND of the NodeMcu and the D0 pin is connected to the D4 terminal.

INTERFACING L2C DISPLAY WITH NODEMCU

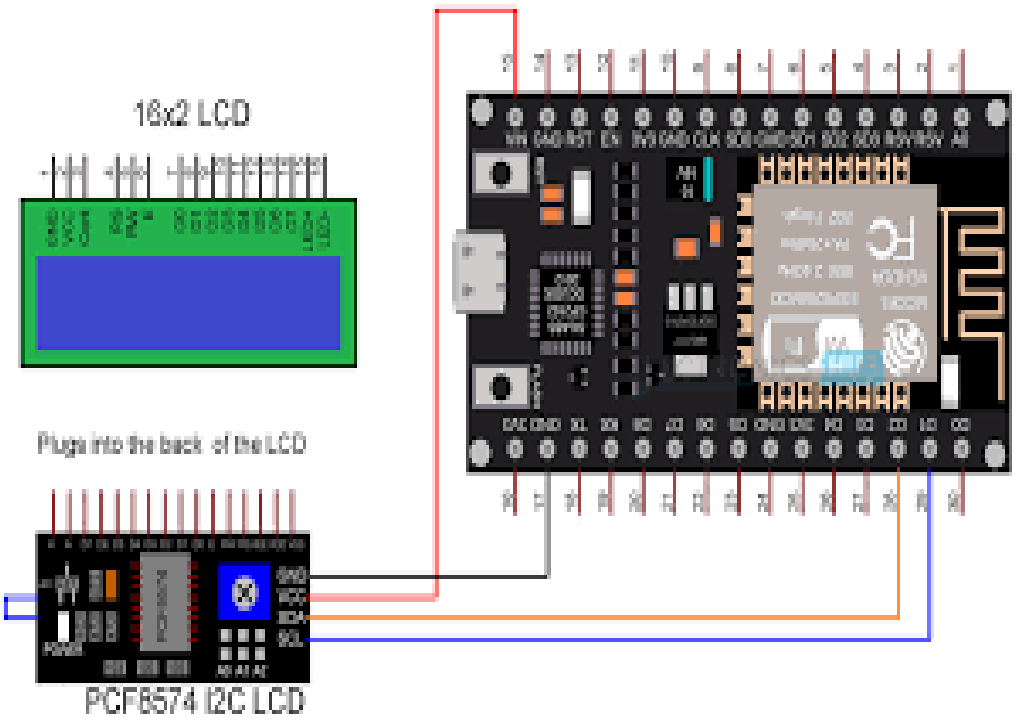


Figure 6 Interfacing with I2C NodeMcu

The outputs from the LDR sensor,DHT 11 sensor, FC37 Rain sensor are displayed at the display of the L2C display. The L2C display updates the user about the real time situation. This module has 4 pins,GND,VCC,SDA and SCL. The D1 and D2 pins are reserved for L2C display,ie, the SDA and SCL pins are connected here respectively. *The* VCC and GND are connected to the Vin and GND of the NodeMcu.

INTERFACING MICRO SERVO MOTOR WITH NODEMCU

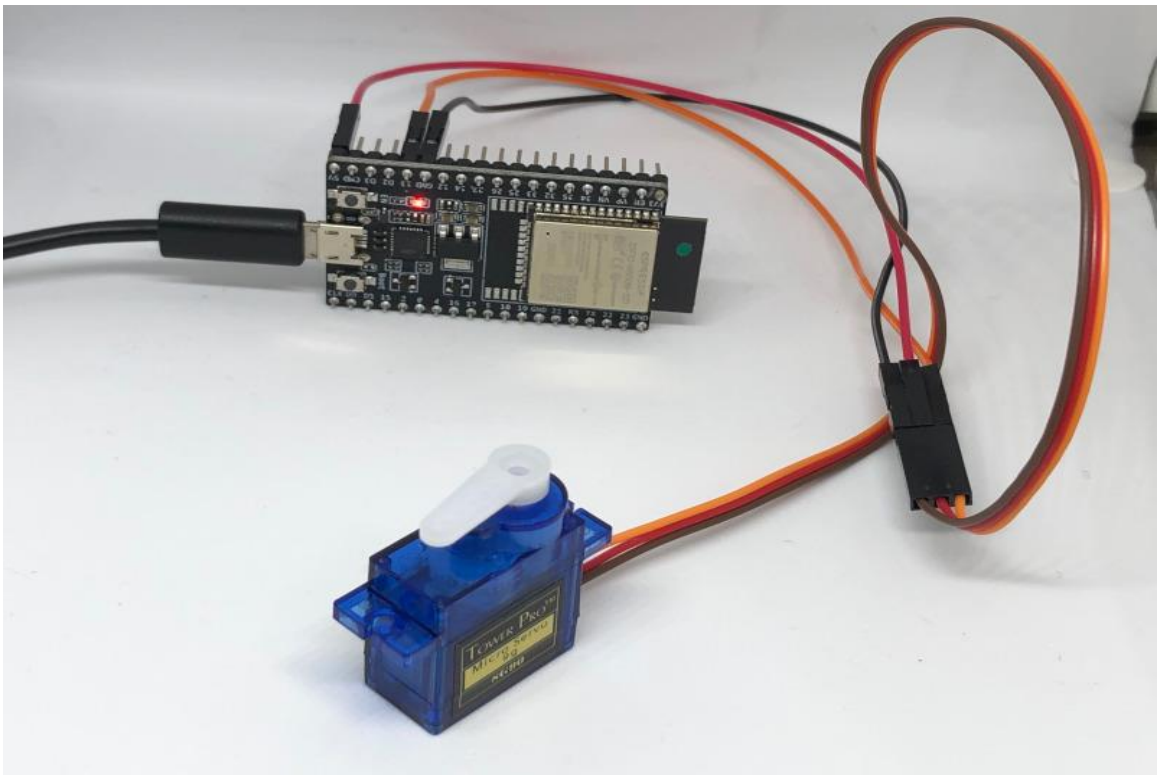


Figure 7 Interfacing Micro servomotor with NodeMcu

The micro servo motor is the main component of this project. This motor is attached at the head of the rain shed. When the rain is sensed the motor rotates the shed to 180degress thereby protecting the object.When no rain is sensed, the motors remains at 0 degree. The micro servo motor has three terminals. The VCC and GND are connected to respective Vin and GND of NodeMCU and the digital pin is connected to the D8 of the NodeMCU.

CHAPTER 7
THE MOBILE APPLICATION- RAIN GUARD

The project can be controlled and monitored by the using a mobile app. This mobile app is being created using a software called ‘BLYNK’. The Blynk application is an open source app making platform which is best for making IOT based project and Automation. It is very easy to link our project to this app and control the same from home.

The first step to create this app is to download ‘Blynk’ from playstore and create an account. After creation of the account, Select create project and enter project name and the micro controller board to be chosen. The app sends us an unique authorization token which must to added to our program code to grant access to our project. The wifi name and password must also be provided in the program code. The user has to connect the the mobile wifi with NodeMcu before starting the project.

The next step is to customize our app. Here we created 4 guage buttons for getting the real time values of temperature, humidity, rain levels and light levels. Virtual pins V0,V1, V2 and V3 are allotted for each of this buttons. A slider named ‘PROTECTION’ is provided which rotates the servomotor from 0 to 180degrees.When the slider is at 0 degree the protection is in OFF position and when its at 180 degrees the protection is at ON position. A notification button is provided at the bottom of the page. When the rain sensor senses rain it immediately sends a notification to the using saying ‘Its raining!!! Turn on protection’. The user has to turn on the protection at this moment.

Some screenshots of the app are shown in the figures below-



Figure 8 Screenshot of app with slider ON

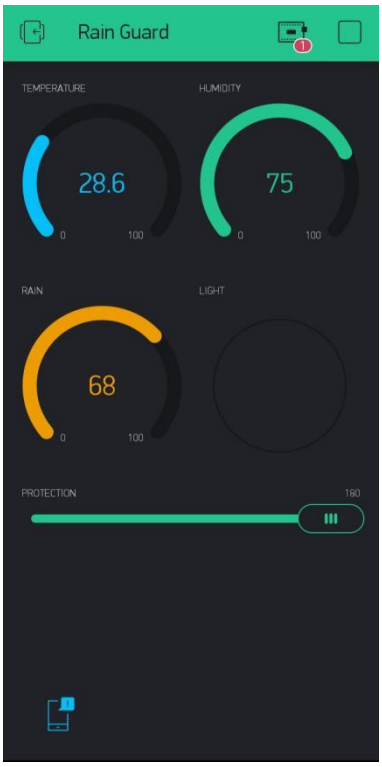


Figure 9 Screenshot of app with slider ON

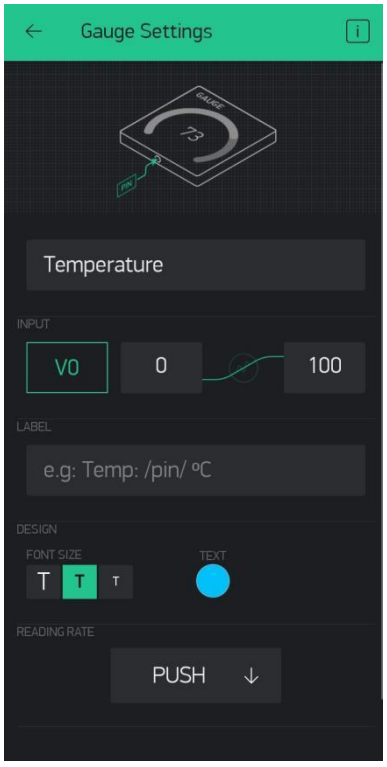


Figure 10 Guage settings of temperature button

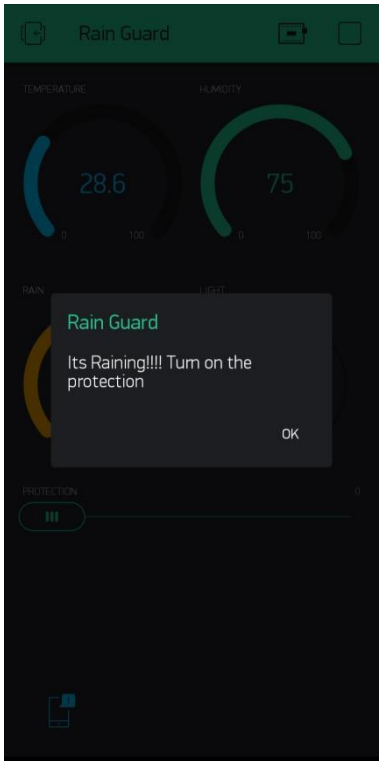


Figure 11 App notification

CHAPTER 8

FLOWCHART

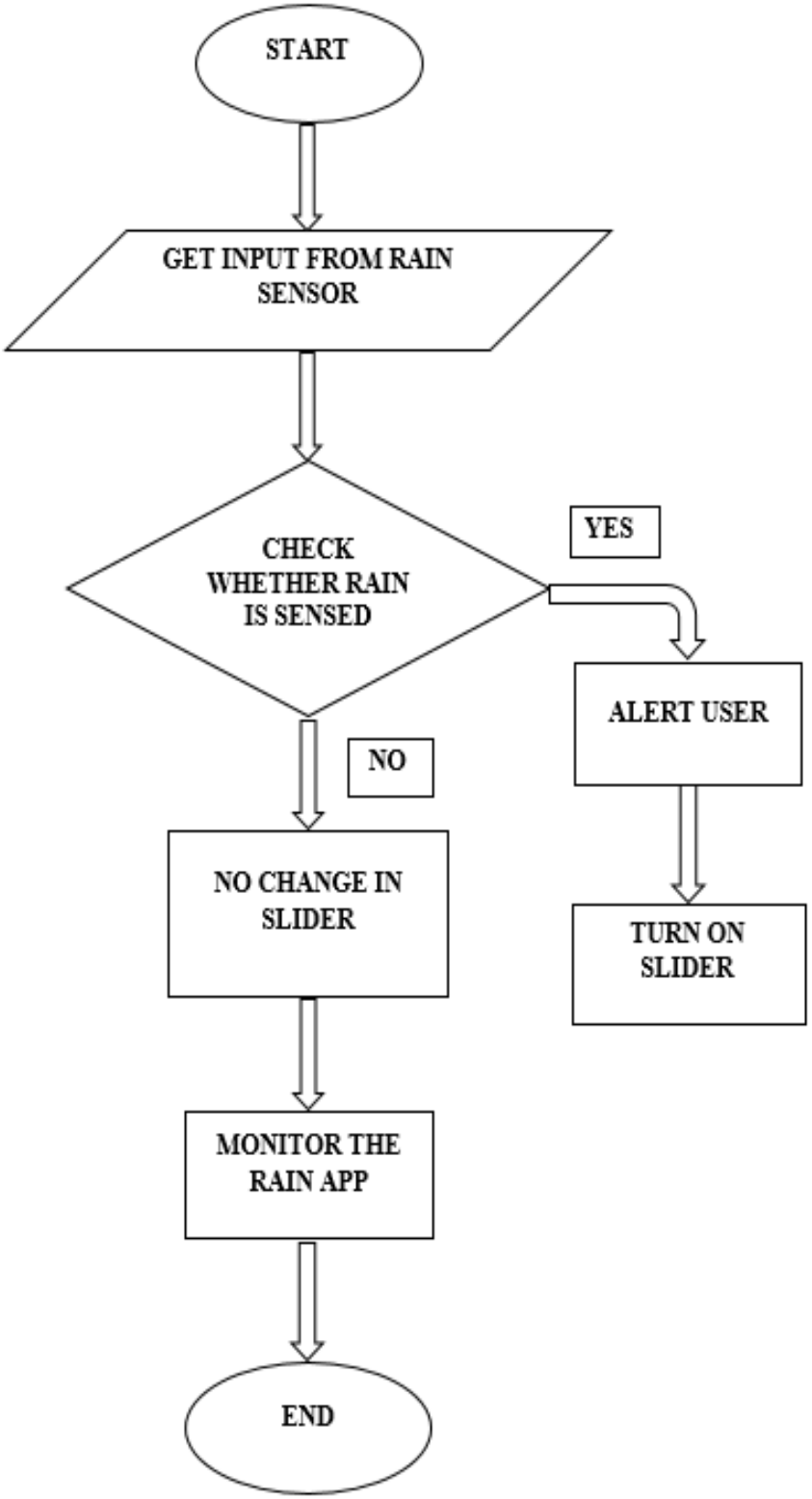


Figure 12 Flowchart of IOTRPS

CHAPTER 9

HARDWARE DETAILS

9.1. NodeMCU ESP8266 Wifi Development Board



Figure 13 NodeMcu ESP8266 Wifi Dev. board

NodeMCU is an open-source firmware and development kit that helps you to prototype or build IoT products. It includes firmware that runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module. The firmware uses the Lua scripting language. It is based on the eLua project and built on the Espressif Non-OS SDK for ESP8266.The firmware is based on the eLua project, and built on the Espressif Non-OS SDK for ESP8266. It uses many open source projects, such as lua-cjson and SPIFFS. Due to resource constraints, users need to select the modules relevant for their project and build a firmware tailored to their needs. Support for the 32-bit ESP32 has also been implemented.

The prototyping hardware typically used is a circuit board functioning as a dual in-line package (DIP) which integrates a USB controller with a smaller surface-mounted board containing the MCU and antenna. The choice of the DIP format allows for easy prototyping on breadboards. The design was initially based on the ESP-12 module of the ESP8266, which is a Wi-Fi SoC integrated with a Tensilica Xtensa LX106 core, widely used in IoT applications.

It’s perfect for IoT applications, and other situations where wireless connectivity is required.

9.2. FC37 Rain sensor



Figure 14 FC37 Rain sensor

A rain sensor is one kind of switching device which is used to detect the rainfall. It works like a switch and the working principle of this sensor is, whenever there is rain, the switch will be normally closed.

The rain sensor module/board is shown in the fig. Basically, this board includes nickel coated lines and it works on the resistance principle. This sensor module permits to gauge moisture through analog output pins & it gives a digital output while moisture threshold surpasses.This module is similar to the LM393 IC because it includes the electronic module as well as a PCB. Here PCB is used to collect the raindrops. When the rain falls on the board, then it creates a parallel resistance path to calculate through the operational amplifier.This sensor is a resistive dipole, and based on the moisture only it shows the resistance. For example, it shows more resistance when it is dry and shows less resistance when it is wet.

This sensor includes four pins which includes, Pin1 (VCC): It is a 5V DC pin; Pin2 (GND): it is a GND (ground) pin; Pin3 (D0): It is a low/ high output pin; Pin4 (A0): It is an analog output pin. In this project, the A0 pin helps to detect the intensity and the D0 pin notifies the using the present updates of rain and whether to turn on the protection or not.

9.3. LDR Sensor Module

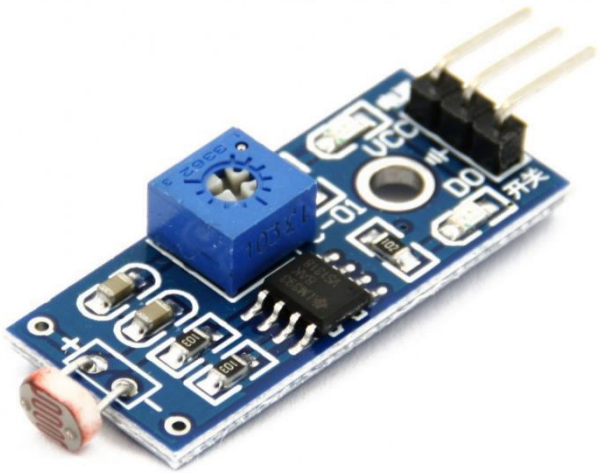


Figure 11 LDR Sensor

LDR sensor module is a low-cost **digital sensor** as well as **analog sensor** module, which is capable to measure and detect light intensity. This sensor also is known as the **Photoresistor sensor**. This sensor has an onboard LDR(Light Dependent Resistor), that helps it to detect light. This sensor module comes with 4 terminals. Where the “DO” pin is a digital output pin and the “AO” pin is an analog output pin. The output of the module goes high in the absence of light and it becomes low in the presence of light. The sensitivity of the sensor can be adjusted using the onboard potentiometer.

The pins of the LDR sensor includes External 3.3V-5V VCC, External GND GND, DO digital output interface, a small plate (0 and 1), AO small board analog output interface.

9.4. DHT11 Senor Module



Figure 12 DHT11 Sensor

DHT11 is a low-cost digital sensor for sensing temperature and humidity. This sensor can be easily interfaced with any micro-controller such as Arduino, Raspberry Pi etc… to measure humidity and temperature instantaneously.

DHT11 humidity and temperature sensor is available as a sensor and as a module. The difference between this sensor and module is the pull-up resistor and a power-on LED. DHT11 is a relative humidity sensor. To measure the surrounding air this sensor uses a thermistor and a capacitive humidity sensor.DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels. The IC measure, process this changed resistance values and change them into digital form.

DHT11 sensor has four pins- VCC, GND, Data Pin and a not connected pin. A pull-up resistor of 5k to 10k ohms is provided for communication between sensor and micro-controller.

9.5. Micro servomotor SG90



Figure 13 Micro Servomotor

Micro Servo Motor SG90 is a tiny and lightweight server motor with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. This component is good to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places. It comes with a 3 horns (arms) and hardware.

This components is the most important functionality in this project. The micro servo motor SG90 is attached to the rain shed. Whenever the user turns on the slider the motor rotates 180 degrees to protect the object from rain.

The micro servo motor has three pins, Red wire- VCC pin, Brown wire-GND pin and the Orange wire- PWM pin.

9.6. I2C Display Module



Figure 14. L2C Module

I2C Module has a inbuilt PCF8574 I2C chip that converts I2C serial data to parallel data for the LCD display.

These modules are currently supplied with a default I2C address of either 0x27 or 0x3F. To determine which version you have check the black I2C adaptor board on the underside of the module. If there a 3 sets of pads labelled A0, A1, & A2 then the default address will be 0x3F. If there are no pads the default address will be 0x27.

The module has a contrast adjustment pot on the underside of the display. This may require adjusting for the screen to display text correctly. The I2C module operates at 5V. The Backlight and Contrast is adjusted by potentiometer in the module. Serial I2C control of LCD display is done using PCF8574. They come with 2 IIC interface, which can be connected by Dupont Line or IIC dedicated cable and are also compatible for 16x2 LCD.

This is another great IIC/I2C/TWI/SPI Serial Interface. With this I2C interface module, you will be able to realize data display via only 2 wires. The pin description of the I2C module consists of 4 pins VCC,GND,SDA,SCL.

9.7. Breadboard

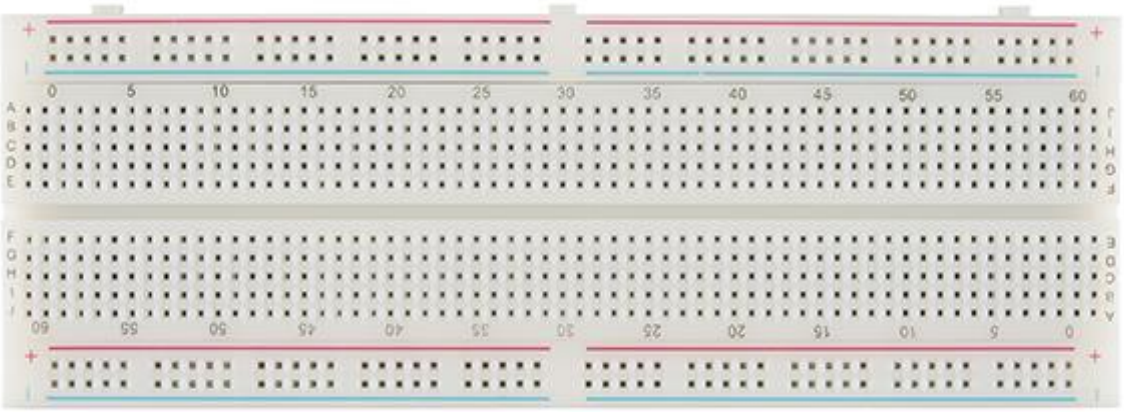


Figure 15. Breadboard

A **breadboard**, or protoboard, is a construction base for prototyping of electronics. Originally the word referred to a literal bread board, a polished piece of wood used when slicing bread.

Because the solderless breadboard does not require soldering, it is reusable. This makes it easy to use for creating temporary prototypes and experimenting with circuit design. For this reason, solderless breadboards are also popular with students and in technological education.

Older breadboard types did not have this property. A stripboard (Veroboard) and similar prototyping printed circuit boards, which are used to build semi-permanent soldered prototypes or one-offs, cannot easily be reused. A variety of electronic systems may be prototyped by using breadboards, from small analog and digital circuits to complete central processing units (CPUs)

9.8. 9V Battery and DC adapter



Figure 16. 9V Battery and DC adapter

A 9V - or Nine-volt - battery is recognizable by its rounded rectangular shape rather than the cylindrical design of other common standard batteries (think AA/AAA/ C/ D, for example). 9V batteries are commonly made of six individual cells that are then enclosed in a singular wrapper to form a block and are available in several chemistry types, with standard 9V batteries typically being either carbon-zinc or alkaline or lithium-iron disulphide batteries.

Typically, a 9V battery is capable of providing the equivalent to six AA cells, however the compact nature of a 9V battery may make it more practical than using multiple smaller cells.

These batteries are particularly useful in long-life applications that require a lot of power due to their high energy capacity.A long performance life also means the battery should be changed at a lower frequency than smaller cells, making them ideal for environments where hygiene is incredibly important. This is why 9V batteries can often be found in medical devices and equipment, such as surgical lighting, data recorders, some patient monitors and even some medical instrumentation.

In this project 9V battery powers up the circuit. The terminals of the battery is connected to the common VCC and GND pins of NodeMCU via a 9v dc adapter.

9.9. Jumper Wires



Figure 17. Jumper Wires

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed.

Fairly simple. In fact, it doesn’t get much more basic than jumper wires.

Jumper wires typically come in three versions: male-to-male, male-to-female and female-to-female. The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things, while female ends do not and are used to plug things into. Male-to-male jumper wires are the most common and what you likely will use most often. When connecting two ports on a breadboard, a male-to-male wire is what you’ll need.

CHAPTER 10

ADVANTAGES, DISADVANTAGES AND APPLICATIONS

ADVANTAGES

Automation is the greatest advantage of the IOT based Rain protection system. The human involvement can be eliminated to an extent. The user can controlled the project from his home via the mobile app and protect the material whenever rain is sensed. The user gets time to pre plan his actions in case of heavy rain by monitoring various parameters provided in the app. More importantly a bulk amount of raw materials can be safeguarded by this project and the margins of errors are also negligible. The components and sensors used in the project is quite cost effective and provide accurate results. By looking into the real time parameters, the user can predict the intensity of the rain to some extent and act according to the situation

DISADVANTAGES

The main disadvantage of this project is its Wifi range. Wifi is the interface between the project and mobile app. Therefore the user must be within the range 150-300 feet from the project.If the user goes beyond this then they might get disconnected from the project. This issue can be resolved by adding GSM in place of Wifi. The project is only meant for protection from rain therefore human involvement is necessary to carry the material indoors after dusk.

APPLICATIONS

The project holds a lot of important and beneficial applications which can be adopted in various areas. It can be used for home automation i.e, for closing and opening doors and windows when it starts to rain. We can also use this project for agricultural purposes, so that it automatically turns off the irrigation system and even notify/alert the user. In automobile sector, we can develop smart wiping system that works when rain is sensed. We can use the project at restaurants located on the rooftop so that when it rains the shade automatically expands and covers the rooftop and we can enjoy our food without hesitation. Clothes can also be protected from sudden rain by provided a shed which rotates after sensing the rain.

CHAPTER 11

CONCLUSION AND FUTURE SCOPE

At the end of this project, a system was designed which could reduce the problem of frequent human involvement and man power to some extent. The project can prove to be very beneficial for industries and for farmers who need to protect their raw materials from sudden rains. Through this project we emphasis the importance of IOT based projects and how Automation can significantly safe time and improve the quality of life of people. The farmers using this project may get a good exposure to the latest trends in technology and the functionalities of Internet of thing.

Other than the factors mentioned above, the project has very good scope in other various sectors like automobile, irrigation systems and household chores. With slight modifications in the project, we can create an automatic car wiping system when can automatically function after sensing rain.The agricultural field can also benefit from this project be creating a smart irrigation system for crops. This project is also very helpful in households as it can be very using protect the clothes from getting wet in rains. As the technology is being giving more to AI robots, this project can be used in such robots to completely eliminate man power and completely controlling the process with help of this robot. We expect that this project of our can makes the life simple and easier for our next generation.

CHAPTER 12

REFERENCES

1. A.Z. Alkar et al, “An Internet based wireless home automation system for multifunctional devices”, 2005 IEEE Transactions on Consumers Electronics,2005, pp.1169-1174, doi:10.1109/TCE.2005.1561840

2. N.J.C Libatique et al, “Design of a tropical rain- Disaster alarm system: A new approach based on wireless sensor networks and acoustic rain rate measurements”,2009 IEEE Instrumentation and Measurement Technology Conference, 2009, pp. 1337-1340,doi:10.1109/MTE.2009.5168663

3. E. M. Trono et al., "Rainfall monitoring using acoustic sensors," TENCON 2012 IEEE Region 10 Conference, 2012, pp. 1-6, doi: 10.1109/TENCON.2012.6412284

APPENDIX

```
#include <LiquidCrystal_I2C.h>
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>
#include <Servo.h>

Servo servo;

LiquidCrystal_I2C lcd(0x27, 16, 2);
 DHT dht(D3, DHT11); //(sensor pin,sensor type)
BlynkTimer timer;
BlynkTimer timer1;

char auth[] = "anZq33IxSdKaZGTm9VYuwFRgfPqyVaw3"; //Enter the Auth code which was send by Blink
char ssid[] = "realme Narzo 20";  //Enter your WIFI Name
char pass[] = "anshul2001";  //Enter your WIFI Password

void weather() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();
  int r = analogRead(A0);
  bool l = digitalRead(D4);
```

```
r = map(r, 0, 1023, 100, 0);
if (isnan(h) || isnan(t)) {
  Serial.println("Failed to read from DHT sensor!");
  return;
}

Blynk.virtualWrite(V0, t);  //V0 is for Temperature
Blynk.virtualWrite(V1, h);  //V1 is for Humidity
Blynk.virtualWrite(V2, r);  //V2 is for Rainfall

if (l == 0) {
  WidgetLED led1(V3);
  led1.on();
  lcd.setCursor(9, 1);
  lcd.print("L :");
  lcd.print("High");
  lcd.print(" ");
} else if (l == 1) {
  WidgetLED led1(V3);
  led1.off();
  lcd.setCursor(9, 1);
  lcd.print("L :");
  lcd.print("Low");
  lcd.print(" ");
}
```

```
 lcd.setCursor(0, 0);
  lcd.print("T :");
  lcd.print(t);

  lcd.setCursor(0, 1);
  lcd.print("H :");
  lcd.print(h);

  lcd.setCursor(9, 0);
  lcd.print("R :");
  lcd.print(r);
  lcd.print("   ");

}
 int flag=0;
void notifyRain()
{
  int isButtonPressed = digitalRead(D7);
  if (isButtonPressed==0 && flag==0) {
    Serial.println("Its Raining");
    Blynk.notify("Its Raining!!!! Turn on the protection");
    flag=1;
  }
  else if (isButtonPressed==1)
  {
    flag=0;
  }
```

```
}
void setup() {
  Serial.begin(9600); // See the connection status in Serial Monitor
  lcd.init();
  lcd.backlight();
  Blynk.begin(auth, ssid, pass);
  servo.attach(15); // NodeMCU D8 pin
  dht.begin();
  // Setup a function to be called every second
  timer.setInterval(10L, weather);
  pinMode(D7 ,INPUT);
  timer1.setInterval(1000L,notifyRain);
}
void loop() {
  Blynk.run(); // Initiates Blynk
  timer.run(); // Initiates SimpleTimer
  timer1.run();
}
BLYNK_WRITE(V4)
{
servo.write(param.asInt());
}
```